



1 Decay Scheme

³²P decays 100% by β^- transition to the ground state of ³²S.
Le ³²P se désintègre exclusivement par émission β^- vers le niveau fondamental du ³²S.

2 Nuclear Data

$T_{1/2}({}^{32}\text{P})$: 14,273 (7) d
 $Q^-({}^{32}\text{P})$: 1710,661 (40) keV

2.1 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^-_{0,0}$	1710,661 (40)	100	Allowed	7,90673 (22)

3 Atomic Data

3.1

ω_K : 0,0804 (19)
 $\bar{\omega}_L$: 0,00093 (18)
 n_{KL} : 1,807 (7)

3.1.1 X Radiations

		Energy (keV)	Relative probability
X _K	Kα ₂	2,30667	50,53
	Kα ₁	2,30787	100
	Kβ ₁	2,4641	9,33
X _L	Lℓ	0,1513	
	Lη	0,1531	
	Lβ	0,225 - 0,2272	

3.1.2 Auger Electrons

		Energy (keV)	Relative probability
Auger K			
	KLL	1,985 - 2,119	100
	KLX	2,227 - 2,300	12,4
	KXY	2,440 - 2,456	0,38
Auger L		0,1326 - 0,2132	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
$\beta^-_{0,0}$	max:	1710,661 (40)	} 100
	avg:	693,306 (18)	

5 Main Production Modes

$\left\{ \begin{array}{l} {}^{31}\text{P}(\text{n},\gamma){}^{32}\text{P} \quad \sigma : 172 \text{ barns} \\ \text{Possible impurities: } {}^{33}\text{P} \text{ (thermal neutrons)} \end{array} \right.$

$\left\{ \begin{array}{l} {}^{32}\text{S}(\text{n},\text{p}){}^{32}\text{P} \quad \sigma : 65 \text{ barns} \\ \text{Possible impurities: } {}^{33}\text{P}, {}^{35}\text{S} \text{ (fast neutrons)} \end{array} \right.$

$\left\{ \begin{array}{l} {}^{35}\text{Cl}(\text{n},\alpha){}^{32}\text{P} \\ \text{Possible impurities: } {}^{36}\text{Cl}, {}^{34}\text{P} \end{array} \right.$

Others: ${}^{29}\text{Si}(\alpha,\text{p}){}^{32}\text{P}$; ${}^{30}\text{Si}({}^3\text{He},\text{p}){}^{32}\text{P}$; ${}^{31}\text{P}(\text{d},\text{p}){}^{32}\text{P}$; ${}^{34}\text{S}(\text{d},\alpha){}^{32}\text{P}$.

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